Parisa Dehghan

R&D Specialist — IoT | Machine Learning | Scientific Computing

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PROFESSIONAL SUMMARY

Interdisciplinary R&D Specialist combining physics, computational science, machine learning, IoT systems, and experimental micro/nano fabrication. M.Sc. in Particle Physics and Field Theory (Shahid Beheshti University) with hands-on experience across the full sensing stack: MEMS simulation and fabrication like 1. silicon AFM-sensor fabrication (UV lithography, RIE, PVD, SEM/AFM metrology) and COMSOL multiphysics simulation 2. microfluidic mask designs; LoRa/MQTT/Node-RED IoT web application with Node.js + SQL back ends; ML deployment including TinyML (quantization, pruning) on Raspberry Pi. Proficient in Python, C/C++, SQL, and Fortran for numerical simulation, HPC, and scientific ML.

TECHNICAL SKILLS

- **Programming:** Python (TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenCV), C/C++ (algorithms, HPC), Fortran, MATLAB, C#/.NET, SQL (SQL Server, Oracle, SQLite/MySQL).
- **Machine Learning & Data Science:** Classification, regression, clustering (KMeans), RNN / stacked and Bidirectional LSTM / CNN / autoencoders / GCN, TinyML quantization + pruning (50% smaller at >90% accuracy), topological data analysis (GUDHI, Dionysus, JavaPlex, persistent homology, Mapper), feature engineering, Naive Bayes, NLP.
- **Scientific Computing & Simulation:** Numerical QFT simulation, Monte Carlo / Metropolis, stochastic differential equations, SPH, CFD (STAR-CCM+ macros), COMSOL multiphysics, Rivet in Docker, HPC.
- **IoT & Embedded Systems:** LoRa wearables simulation (Node-RED), MQTT, Node.js REST/MQTT services, Raspberry Pi + SSH, sensor networks, real-time telemetry (GPS, blood pressure, SpO2, fall / water-immersion detection), edge ML.
- **Micro/Nano Fabrication & Characterization:** AFM-sensor flow on Si wafers (UV lithography, RIE, wet etching, PVD Au films, high-aspect-ratio tips), PDMS / microneedle masters, hydroxyapatite + PVP formulation R&D, masks (LayoutEditor, CorelDraw), SEM / AFM / optical microscopy, CATIA, KOH-etch + TFET simulation.

PROFESSIONAL EXPERIENCE

R&D Specialist — Intelligent Micronano Sensors (Lotus IMNS), Tehran *May 2024 – September 2026*

Micro/nano fabrication + multiphysics simulation for AFM sensors and microneedle patches

- End-to-end silicon AFM probes: mask layout through UV lithography, RIE, wet etching to released high-aspect-ratio tips; ∼500 sensors per wafer; iterative optimization of sharpness and yield.
- Python fabrication-data analysis + COMSOL multiphysics simulation; ML-ready process datasets; PVD gold films; SEM/AFM/optical validation.
- Microneedle patches: PDMS molds from Si masters; hydroxyapatite loading by centrifugation; PVP formulations for transparent, robust, flexible patches; CATIA centrifuge tooling; KOH-etch.

IoT Specialist — Iran IoT Expansion Center, Tehran *May 2025 – May 2026*

Smart health-monitoring IoT web application + IoT instruction

- End-to-end real-time platform ingesting wristband telemetry (GPS, blood pressure, SpO2, fall / water-immersion detection): Node-RED LoRa simulation via MQTT, Node.js REST/MQTT service to SQL (SQLite/MySQL), live HTML/CSS/JS dashboard.
- Delivered 6-week "Intro to IoT" online course for ages 8–12 with hands-on material.

EDUCATION

M.Sc. in Particle Physics and Field Theory — Shahid Beheshti University, Tehran *2021 – 2024*

- GPA 14.13/20: QFT II (18.23/20), Gravity I (17/20), Adv. Quantum Mechanics (16.02/20), Adv. Statistical Mechanics (15.9/20), Geometry and Topology (15/20). Monte Carlo, SDEs, HPC (C++/Python/Fortran).

B.Sc. in Physics — Alzahra University, Tehran *2017 – 2021*

- GPA 15.66/20; 4/4 in Numerical Calculations, Stochastic Processes, Adv. Quantum Mechanics I, Computer Programming (theory + practice).

Diploma in Mathematics and Physics — Motahare High School, Tehran *2013 – 2016*

- GPA 19.38/20.

SELECTED PROJECTS

NeuroTwin — Sensor-Health and Predictive-Maintenance Platform (C# / .NET) *Portfolio flagship*

- Decouples machine vs. sensor faults via physics-motivated features, anomaly detectors, digital-twin residuals; RUL + explainable diagnosis; 7 fault modes; radix-2 FFT, health index, RUL interval; REST API + Docker, 33 tests; NASA

C-MAPSS / CWRU hooks.

MEMS Simulation Application — TypeScript / React *Portfolio flagship*

– Browser-based fabrication simulator: lithography, RIE, BHF, KOH, oxidation on both wafer faces; plane-dependent Hamilton–Jacobi KOH etch front forms (111) walls, V-grooves, undercut, tips; AFM tip height within measured range; CAD-style constraint mask sketcher; GDSII import; 3D views.

Physics-Informed Symbolic Regression (pisr) — Python / PyTorch *Portfolio flagship*

– LLM-free symbolic regression: dimensional analysis + Pi-groups, EFT-style bases, sparse regression, 27k-parameter tree scorer; exact recovery of canonical laws; MDL selection; AI-Feynman / PMLB + noise sweep.

Edge AI: Environmental Monitoring (Raspberry Pi) + TinyML Image Classification *Edge AI*

– Real-time sensor analytics on Raspberry Pi; quantized + pruned classifier 50% smaller with >90% accuracy.

Bubble Rise Simulation — Python SPH + STAR-CCM+ *CFD*

– Python SPH model + automated STAR-CCM+ macro for micron bubble-rise dynamics; scaled CFD, theory comparison, auto export.

TFET Device Simulator — Interactive web app *Semiconductor*

– p-i-n Si TFET transfer + band diagrams with COMSOL script generation.

Centrifuge Rotor — CATIA V5 Mechanism *Mechanical design*

– 4-place swinging-bucket rotor ("+" header; 0–90 deg per $\tan\theta = \omega^2 r/g$); symmetries, hinge constraints, animation.

Graph Convolutional Networks — Presentation *Machine learning*

– Authored paper on GCNs in complex systems.

Ising Phase Transition — Metropolis Monte Carlo *Statistical physics*

– Metropolis Monte Carlo with scrambling for 2D Ising transitions. Code: github.com/Parisadghn/confusion-scheme.

Topological Data Analysis — GUDHI / Dionysus / JavaPlex *Applied topology*

– Persistent homology + Mapper on biological / network data.

Deep Learning: Stock Prediction (LSTM/Bi-LSTM) + Fraud Detection (Autoencoder) *Deep learning*

– Stacked/Bi-LSTM forecaster (TensorFlow/Keras); autoencoder flags high-error transactions as fraud.

Particle Accelerator Simulation — Rivet / Docker / C++ *High-energy physics*

– Rivet-in-Docker simulation of high-energy processes (event to analysis pipeline).

Medical Imaging: Breast Cancer + MRI Analysis — Scikit-learn / OpenCV *Health ML*

– Unsupervised early detection of breast cancer; MRI processing for structural differences in brain disorders.

NLP, Markets, Data — Naive Bayes / KMeans / Pandas *Applied data science*

– Snapfood sentiment (Naive Bayes); crypto clustering (KMeans); Hamshahri Persian-corpus engineering. Laptop-price ML: github.com/Parisadghn/laptop-price-ML.

Stochastic Thermodynamics (SDEs) + Algorithms (Knapsack/Huffman/MST) + OOP Apps *Fundamentals*

– SDE simulations for molecular motors / nano-engines; DP knapsack; Huffman (30% smaller); Kruskal MST (20% gain); OOP library / To-Do / calculator.

CERTIFICATIONS AND TRAINING

– SANS SEC505, Network+, Security+ — LIAN Group (Sep 2025)
– Certified Ethical Hacker (CEH) — Maktabkhooneh (Jul 2025)
– Software Test Bootcamp (SQL Server + Oracle) — Behsa (May 2024)
– DESY Remote Summer School 2023 (Python, C++, Fortran)
– Greedy Algorithms / MST / Dynamic Programming — Stanford / Coursera (2023)
– Stochastic Thermodynamics — ICTP (May 2023)
– Applied ML in Python — Michigan / Coursera (Oct 2022)
– TDA Workshop — SBU (Aug 2022)
– Python — Alzahra (Oct 2021)
– C / C# / SQL — SoloLearn.

TOOLS AND LANGUAGES

– **Databases:** SQL Server, Oracle, SQLite/MySQL; Git/GitHub; Node.js, REST/MQTT, HTML/CSS/JS.
– **Design:** 3ds Max, CATIA, LayoutEditor, CorelDraw, InDesign/Illustrator, Lightroom/Photoshop; analog photography.
– **Languages:** Persian (native); English (fluent, TOEFL iBT 93); French (intermediate). **Interests:** ballet, painting, guitar, analog photography.